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Roll No

FT-6002 (CBGS)

B.E. VI Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Nuclear Safety and Radioactive Materials

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Find half life period of radioactive material in terms of decay constant.
b) What are the types of exposure? Discuss them.
c) What are the biological effect of radiation on long time exposure?
2. a) Define Radiation, draw Electromagnetic spectrum and its life examples? Why some isotopes are radioactive, Deduct formula for radioactive half-life linking to decay constant.
b) Discuss HAZMAT.
3. a) Draw and explain the difference between the Geiger Muller counter (GM) and Scintillation Counter (SC) and their uses?
b) What are the Handling and Emergency preparedness plan in Radiation emergency in Nuclear power plant?

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4.
 - a) What are radiation placard and label and their requirements for used in transportation, packaging and storage for radioactive material?
 - b) Discuss Emergency preparedness plan and handling requirements in nuclear power plants for radiation emergency.
5.
 - a) What is Altitude temperature inversion?
 - b) Discuss Iodine absorption and remedies.
 - c) Explain ion-exchange method for treatment of liquid radioactive waste.
6.
 - a) Discuss case study of accident Ishikawa 2011.
 - b) Explain CANDU reactor.
7. Write any case study of nuclear power plant Accident. Discuss the safety conditions required during an accident in Nuclear power plant.
8. Write short notes on : (any two)
 - i) Units of radiation and occupational doses
 - ii) ATWS
 - iii) Role of enriched uranium and moderators
